

Chapter – 4: The World of Metals and Non-Metals

- Yashwant, Anandi – live – village – Rajasthan
- Their school – assigned – project – learn about craftspersons – work with metal
- They decide – visit – local ironsmiths – practice this craft
- Yashwant, Anandi – request grandfather – accompany them
- They are curious – learn – ironsmiths make diff. items – how?
- They interact – elderly craftsperson – Sudarshan Uncle
 - Yashwant – what items you make?
 - Sudarshan – items – daily use – flat pans (*tawa*), buckets (*baltis*), tongs (*chimtas*) – farming tools – spades (*phawras*), axes (*kulhadis*), trowels (*khurpis*), rakes (*jelees*)
 - Anandi – what materials they are made of?
 - Sudarshan – iron metal – make items – wood – make handles – when req. – use coal – furnace – heat iron
- Sudarshan – heating iron – furnace – become red hot – starts beating it – hammer
 - Anandi – why are you beating it?
 - Sudarshan – beating it – shape into axe
 - Anandi – piece of iron – beaten into sheet – can other metals as well?

Properties of Materials

Malleability

- Activity –
 - Collect – waste pieces – copper, aluminium, iron nail, coal, sulphur (pea-sized lump), wood
 - **Observe** – appearance – above items – lustrous? – hard or soft? – record it
 - Place – each item – hard surface – beat them – hammer
 - Record – obs. – flatten or break into pieces
- **Analyse** – record – **identify** – obj. – **lustrous, hard**
- Obj. – copper, aluminium, iron – hard, lustrous – called metals
- Coal, sulphur, wood – not as hard, lustrous as metals
- All metals – hard, solid? – no – some metals – sodium, potassium – **soft** – cut with knife
- Mercury – liquid state – room temp. – seen in thermometers
- Obs. – obj. – copper, iron nail, aluminium – become flat – other obj. – behave diff.
- Prop. – materials – beaten into thin sheet – **malleability** – most metals – this prop.
- Example – thin silver foil on sweets – aluminium foil to wrap food items
- Gold, silver – most malleable metals
- Coal, sulphur – do not show – same behaviour – break into pieces – called **brittle**
- Wood – neither breaks nor flattens – neither malleable nor brittle

Ductility

- Seen – wires – made of metals – copper, aluminium – electrical fittings
- Ornaments – bangles, necklaces, earrings – made of metals
- Metal wires – imp. func. – variety of stringed musical instruments – veena, sitar, guitar, violin
- Prop. – metals – drawn into wires – **ductility**
- Gold – so ductile – 1 gram – drawn into 2 km long wire

Sonority

- Ever heard – sound – metal obj. – spoon, plate, coin – dropped on floor
- Sound – how diff. – coal, wood – dropped on floor
- Activity –
 - Take – few obj. – metal spoon, coin, coal, wood
 - Drop them – one by one
 - Notice – difference in sound
- Obs. – metal spoon, coin – ringing sound – coal, wood – other hand – dull sound
- Prop. – metals – enable them – ringing sound – **sonority** – metals – sonorous in nature

Conduction of heat

- Obs. – cooking vessels – kitchen – made of metals
- Try naming – some metals – used – making – cooking vessels
- Activity –
 - Place glass tumbler on table – fill it – hot water
 - Take – metal spoon, wooden spoon – same size, thickness
 - Immerse – both spoons – hot water – same time – leave – undisturbed
 - Carefully – touch – upper end – each spoon
 - Which spoons – get hotter?
- Notice – metal spoon – hotter than wooden spoon
- Both spoons – immersed in water – same time – same temp.
- This shows – metals spoon – heat transfer – easy – wooden spoon – heat transfer – poor
- Transfer of heat – one pt. to another pt. – material – conduction – materials – transfer heat – **conductors**
- Metals – **good conductors** – wood – **poor conductors**
- This reason – cooking vessels – made of metal – their handles – made of wood, other materials – do not conduct heat

Conduction of electricity

- **Design** – elec. circuit – ‘tester’ – previous chapter
- Repeat – same act. – materials – below – record obs.
- Activity –
 - Collect – aluminium foil, iron nail, dry wood, sulphur (pea-sized), copper wire, coal, stone, rubber eraser, nylon rope
 - **Predict** – which material – bulb glow – which material – bulb – do not glow
- Obj. – aluminium, iron, copper – bulb glow – sulphur, coal, wood, stone, eraser, nylon – bulb do not glow
- Materials – allow electricity – pass through – **good conductors of electricity**
- Materials – prevent electricity – pass through – **bad conductors of electricity**
- Plastic covering – screwdrivers, rubber gloves – protect electrician – electric shock - these materials – poor conductors

Effects of Air and Water in Metals: Iron

- Obs. – iron obj. – brown deposits – left in open – few days – which conditions?
 - Comes in contact – dry air only
 - Comes in contact – water only
 - Comes in contact – air and water both

- Activity –
 - Take – few shining iron nails – if using old nails – scrub them – sandpaper
 - Take – 3 clean, dry glass bottles – tight cap – label them – A, B, C
 - Take – 3 iron nails – tie thread to them
 - Place – 1 iron nail – silica gel – bottle A – tighten the cap
 - Silica gel – makes air dry
 - Used – small pouches – shoe boxes, water bottles
 - Place – 1 iron nail – glass bottle B – tighten the cap
 - Pour freshly boiled, cooled water – remove dissolved gases – dip iron nail completely
 - Pour – layer of oil – remove contact with air
 - Place – 1 iron nail – bottle C – pour water – dip iron nail half – do not cap it
 - Iron nail – contact – water, air
 - Place – glass bottles – undisturbed – room temp. – obs. – 8-10 days
 - Record – obs.
- Bottles – A, B – iron nails – no sign – brown deposit OR discoloration
- Bottle – C – iron nails – presence – brown deposit
- Indicate – presence – both air, water – essential – brown deposits
- Moist air – responsible – development – brown deposits – iron obj.
- Brown deposits – rust – process – forming rust – iron obj. – **rusting**
- Many other metals – discoloration – kept in atm.
- Green coating – forms on copper obj. – black coating – silver obj.
- Gradual deterioration – metal surface – cause – air, water – **corrosion**
- Rusting of iron – serious problem – enormous amt. – spent – replace, repair – damages
- Rusting of iron – prevented – several methods – painting, oiling, greasing, protective layer of zinc metal (galvanization)

Effect of Air and Water on Other Metals

- Activity –
 - Take – Magnesium ribbon – 3-4 cm long – clean it – rub – sandpaper
 - Hold it – pair of tongs – ignite – other end – spirit lamp, candle
 - Let the magnesium burn – obs.
 - Obs. – magnesium ribbon – burns – dazzling white flame – changes into white powder – collect it
 - This powder – magnesium oxide – formed – reaction of magnesium and oxygen
 - Add drops – warm water – stir well – check nature
 - Check – solution – magnesium oxide – acidic, basic, neutral
- Obs. – red litmus – changes colour to blue – hence – basic in nature
- Generally – oxides of metals – basic
- Sodium – metal – kept in kerosene – reacts – vigorously – water, oxygen – lots of heat generated

Substances that Behave Differently from Metals in Air and Water

- Activity –
 - Take – small amt. – powdered sulphur in deflagrating spoon – long-handle spoon
 - Heat it – sulphur starts burning – introduce spoon – gas jar – tighten the lid – ensure – gas does not escape

- After 3-4 minutes – remove lid – remove spoon – add water – small qty. – re-place the lid – shake – dissolve gas
 - Check solution – acidic, basic, neutral
- Obs. – blue litmus – changes colour to red – hence – acidic in nature
- Burning sulphur – air (oxygen) – sulphur dioxide gas – formed – dissolve in water – sulphurous acid – formed
- Activity –
 - Take – sulphur powder – glass tumbler
 - Add – water – small amt. – obs.
- Notice – no reaction – between – sulphur, water
- Substances – sulphur, phosphorus, etc. – behave diff. than metals – air, water
- Phosphorus – kept in water – catches fire – exposed to air
- These substances – usually soft, dull – neither malleable nor ductile – not sonorous – also poor conductors of heat, electricity
- These substances – **non-metals** – their oxides – acidic in nature
- Some other non-metals – oxygen, hydrogen, nitrogen, carbon, etc.
- Not to confuse – materials – plastic, glass, wood, rubber, paper – not classified – metals, non-metals – as – not elements

Are Non-metals Essential in Everyday Life?

- Obs. – many metals – day-to-day life – distinct properties – lustrous nature, heat, electrical conductivity, high strength
- However – non-metals – also imp.
- Oxygen – we breathe in – very imp. – survival
- Carbon – essential – building block – all life forms – key component – proteins, fats, carbohydrates
- Nitrogen – manufacturing – fertilisers, chemicals – essential – plants' growth
- Chlorine – non-metal – commonly-used – water purification
- Iodine solution – applied on wounds – antiseptic